Maharashtra Education Society's
Institute of Management and Career Courses (IMCC), Pune
AUTONOMOUS

[CGPA-3.32] in Cycle 1

Academic Year 2026-27
FY MCA- SEM I
IT12 Data Structure and Algorithms
Practical Assignment1

Date: 15/10/2026 **Marks: 40**
Roll No: **Name:**

Create a Singly Linear Linked List with following operations

1. Create Linked List (append)
2. Traverse and print the node values
3. Insert node at a specific position
4. Delete alternate nodes in the linked list

```
5. class Node:
6.     def __init__(self, value):
7.             self.data = value
8.             self.next = None
9. class LinkedList:
10.      def __init__(self):
11.                  self.head = None
12.      def append(self, new_node):
13.              if self.head == None:
14.                    self.head = new_node
15.              else:
16.                     temp = self.head
17.                     while(temp.next):
18.                            temp = temp.next
19.                     temp.next = new_node
20.
21.#2 Traverse and print teh node values
22.      def print(self):
23.       temp = self.head
24.       while (temp):
25.              print(temp.data)
26.              temp = temp.next
27.
28.#3 Insert Node at a specific position
29.      def insert(self,new_node, pos):
30.       if pos == 1:
31.              new_node.next = self.head
32.              self.head = new_node
```

```
                else:
                p = 1
                temp = self.head
                while(p!=pos-1):
                        temp = temp.next
                        p += 1
                new_node.next = temp.next
                temp.next = new_node

#4 Delete Alternate nodes in the linked list
        def delete(self):
         temp = self.head
         while temp and temp.next:
                temp.next = temp.next.next
                temp= temp.next

list1 = LinkedList()
n1 = Node(10)
n2 = Node(20)
list1.append(n1)
list1.append(n2)

list1.append(Node(30))
list1.append(Node(40))
list1.append(Node(50))
print("Original Linked Lsit: ")
list1.print()

list1.insert(Node(22), 3)
print("Linked List After Insertion : ")
list1.print()

list1.delete()
print("Linked List after deleting nodes: ")
list1.print()
```

```
Problems  Output  Debug Console  Terminal  Ports                bash

ADMIN@PC76 MINGW64 ~/2601106_PalakKolambe
$ python D.py
Original Linked Lsit:
10
20
30
40
50
Linked Lsit After Insertion :
10
20
22
30
40
50
Linked List after deleting nodes:
10
22
40
```